with Spring Boot

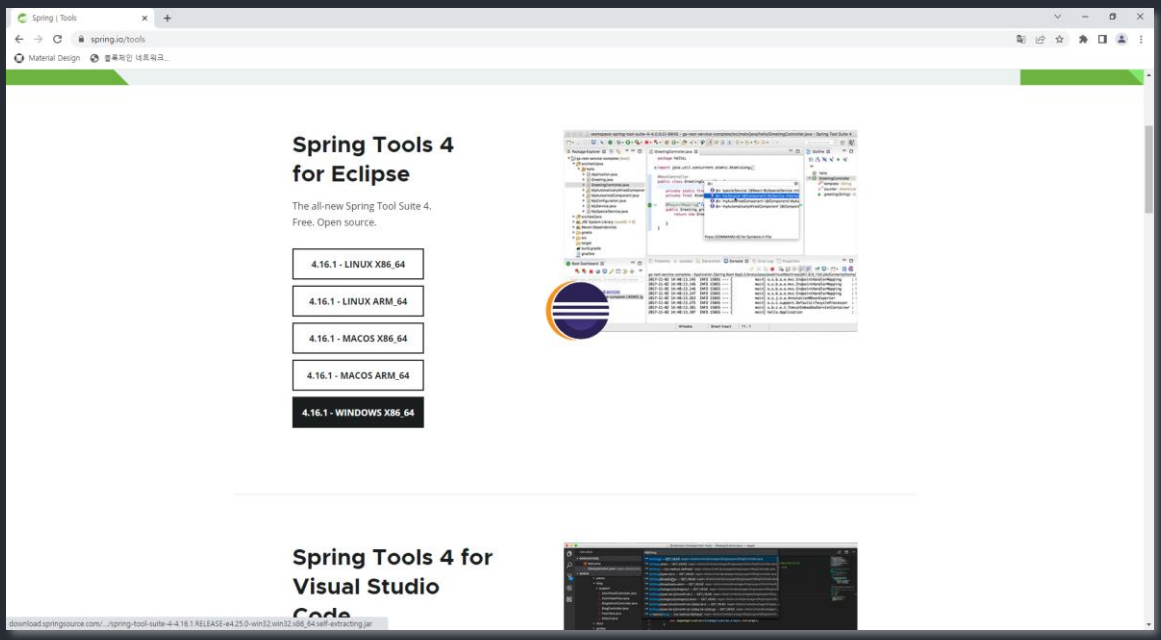
권기웅

1. Spring Boot Back-end

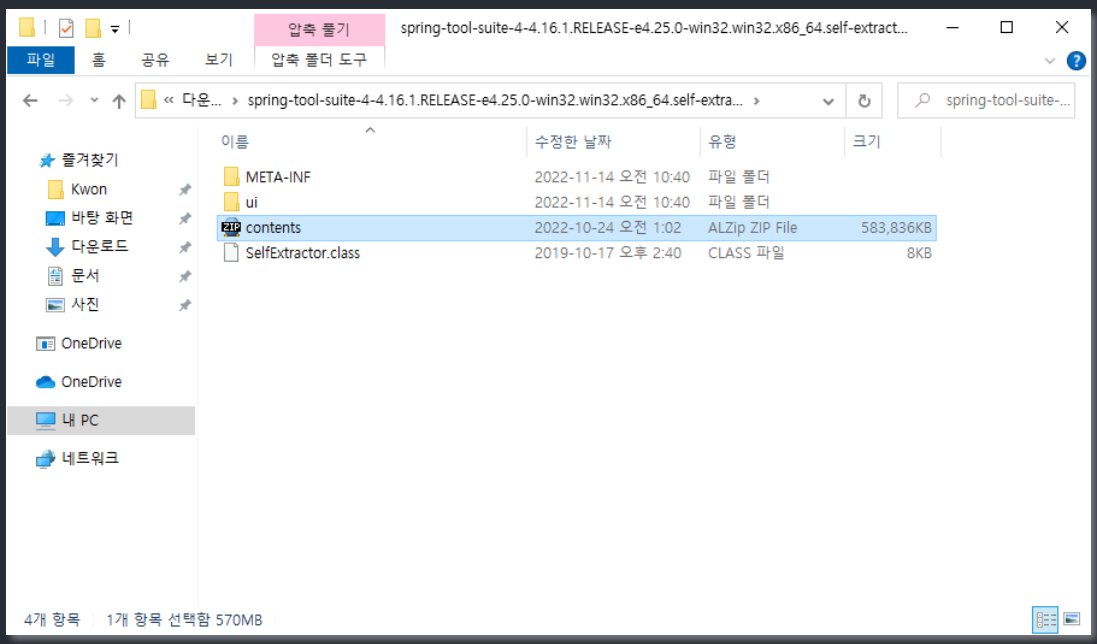
1.1 환경설정

1.1.1 STS

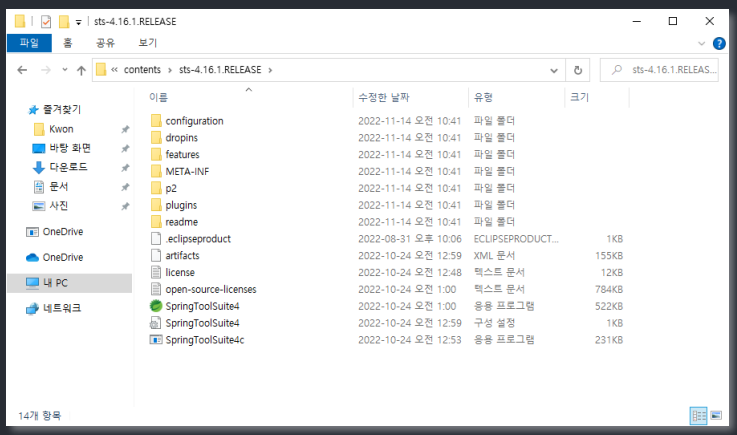
https://spring.io/tools 에서 받아서



.jar 지만 압축 풀고

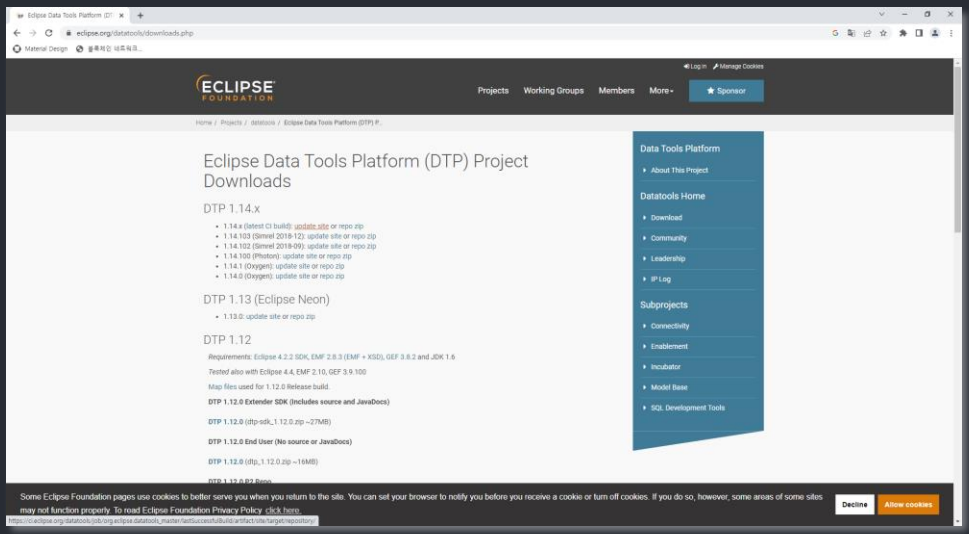


contents.zip 도 풀면 EclipseIDE 나옴

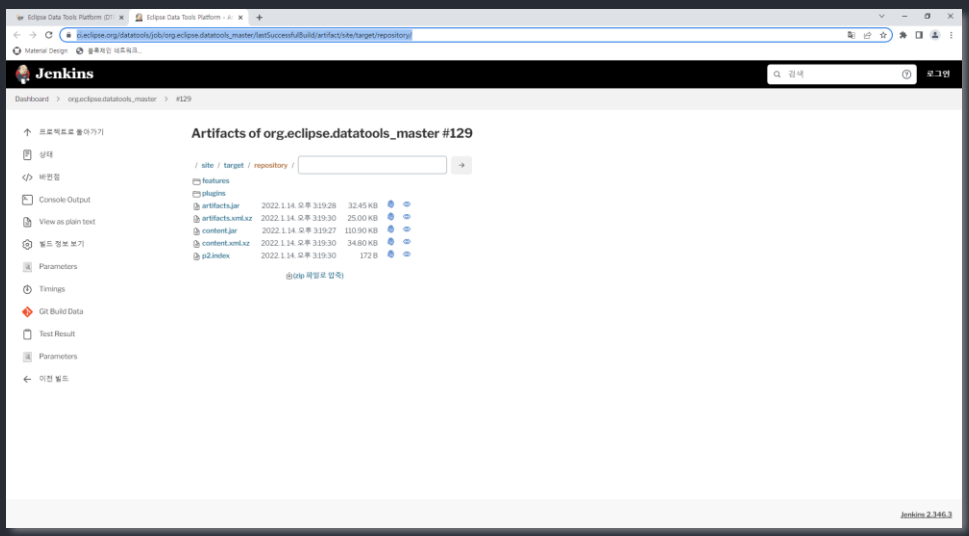


1.1.2 DTP

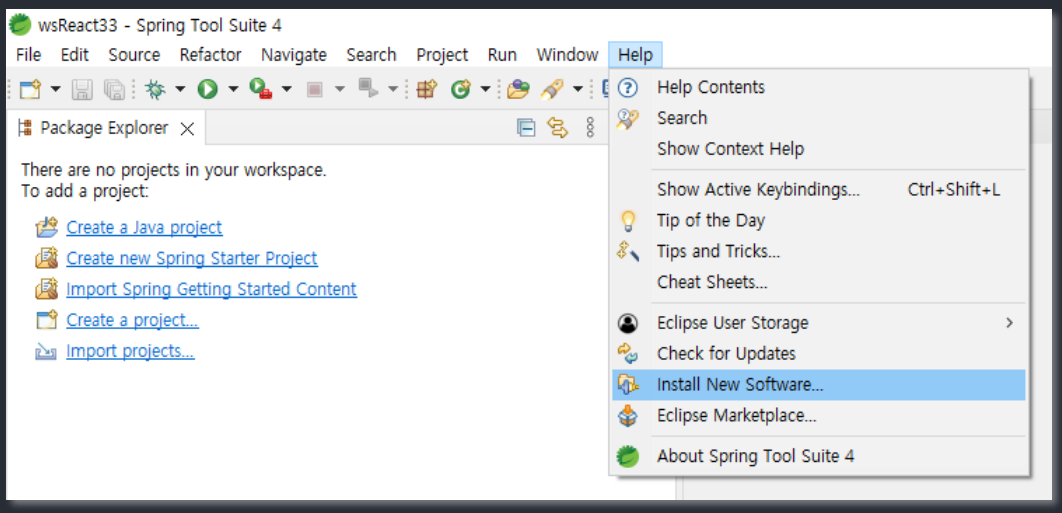
https://www.eclipse.org/datatools/downloads.php

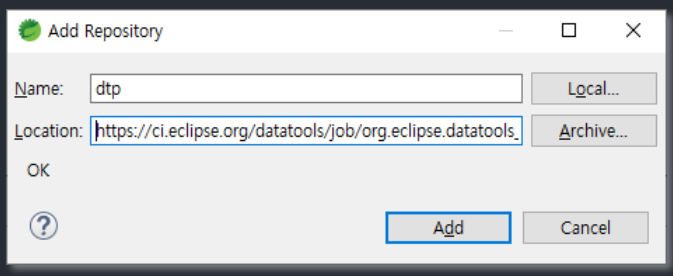


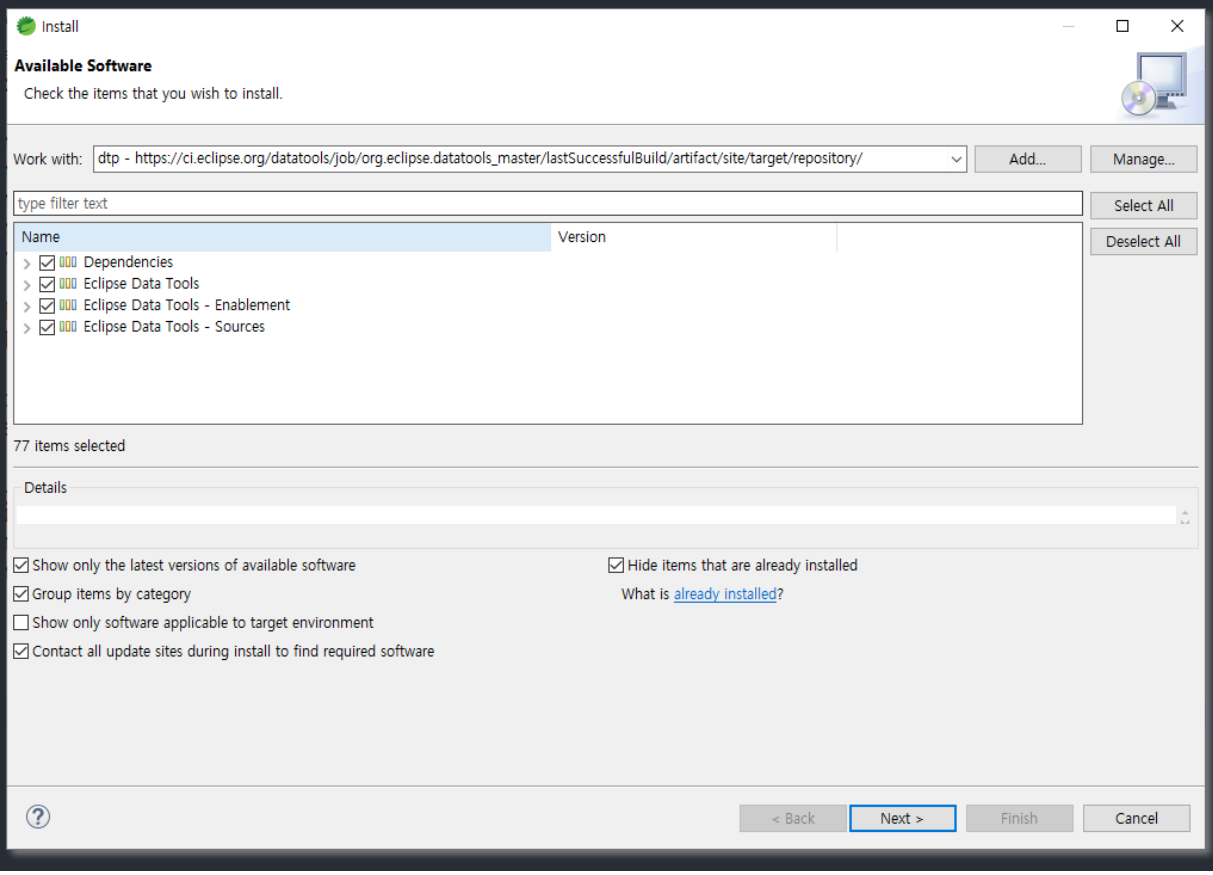
주소 복사해서



설치


wsReact33 - Spring Tool Suite 4
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
There are no projects in your workspace.
To add a project:
Create a Java project
Create new Spring Starter Project
Import Spring Getting Started Content
Create a project...
Import projects...
Help Contents
Search
Show Context Help
Show Active Keybindings... Ctrl+Shift+L
Tip of the Day
Tips and Tricks...
Cheat Sheets...
Eclipse User Storage
Check for Updates
Install New Software...
Eclipse Marketplace...
About Spring Tool Suite 4

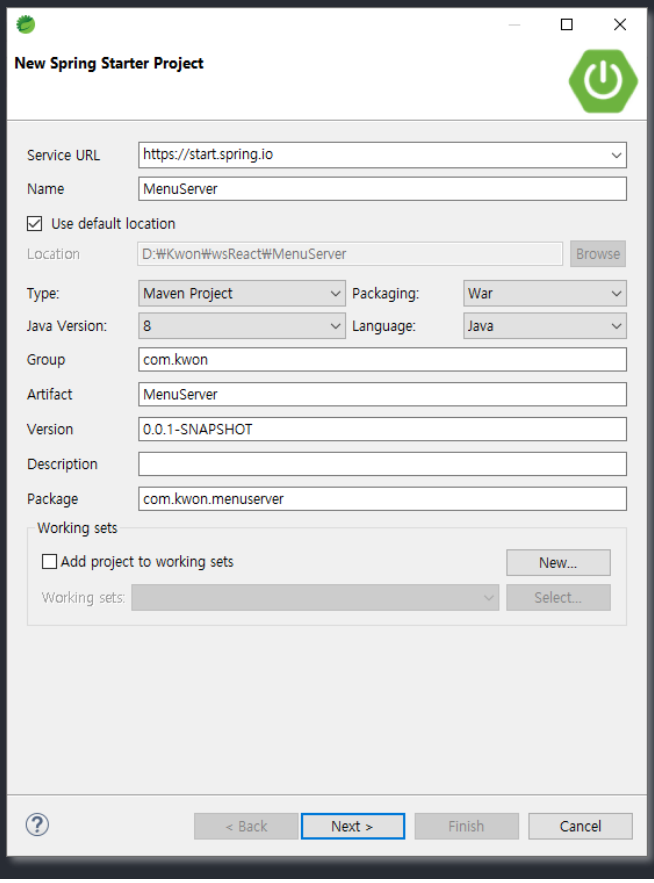

Add Repository
Name: dtp
Local...
Location: https://ci.eclipse.org/datatools/job/org.eclipse.datatools
Archive...
OK
Add
Cancel


Install
Available Software
Check the items that you wish to install.
Work with: dtp - https://ci.eclipse.org/datatools/job/org.eclipse.datatools_master/lastSuccessfulBuild/artifact/site/target/repository/
Add...
Manage...
type filter text
Select All
Deselect All
Name
Version
Dependencies
Eclipse Data Tools
Eclipse Data Tools - Enablement
Eclipse Data Tools - Sources
77 items selected
Details
Show only the latest versions of available software
Group items by category
Show only software applicable to target environment
Contact all update sites during install to find required software
Hide items that are already installed
What is already installed?
< Back
Next >
Finish
Cancel

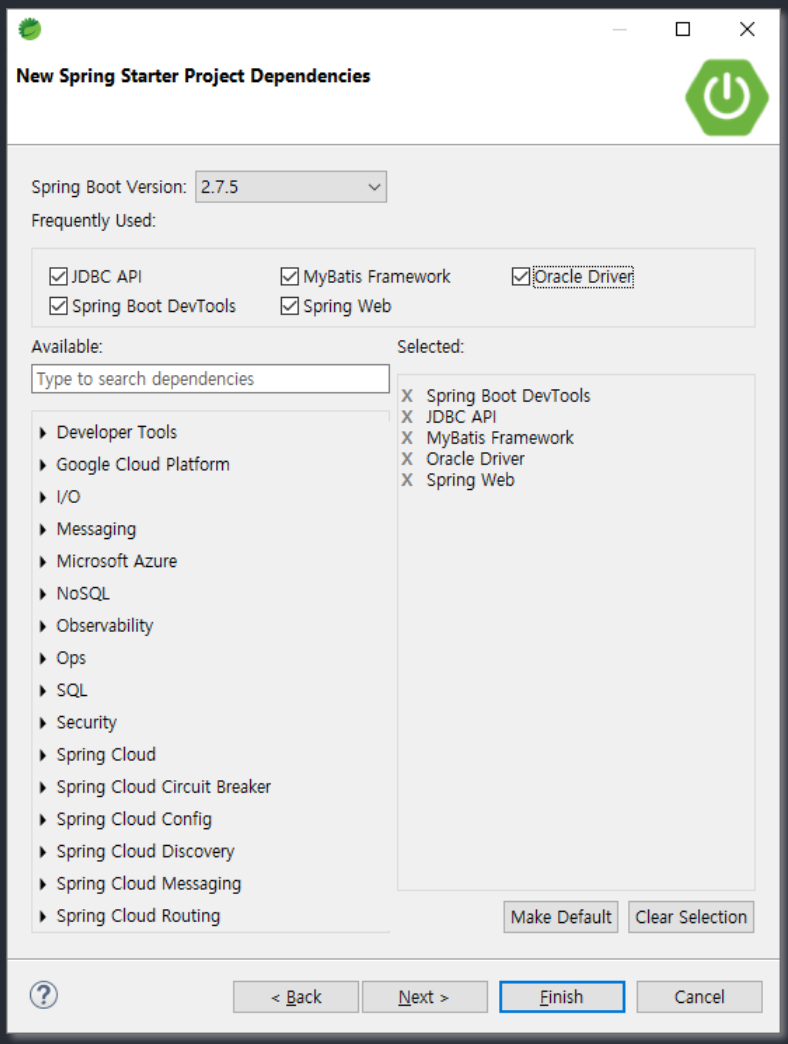
1.2 프로젝트

1.2.1 프로젝트 만들기

Spring Starter Project

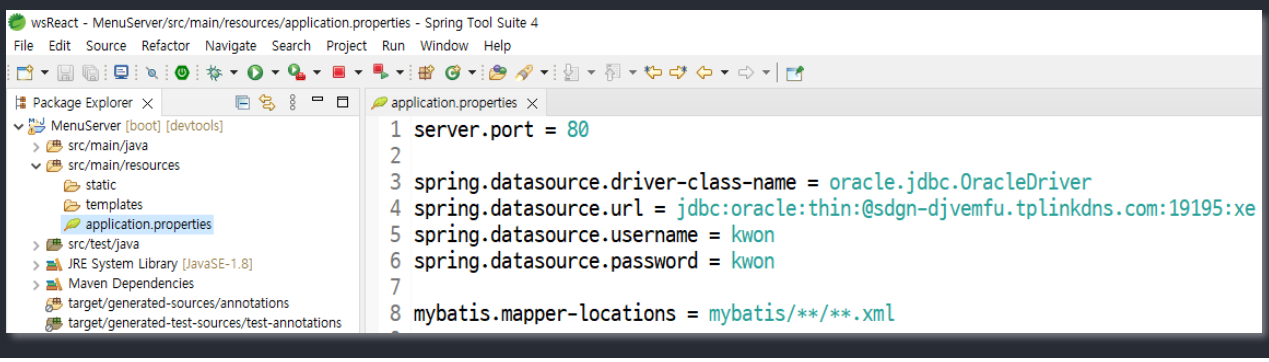


1.2.2 써먹을 것들 선택



1.2.3 프로젝트 설정 파일

src/main/resources/application.properties



1.2.4 DB

db.sql

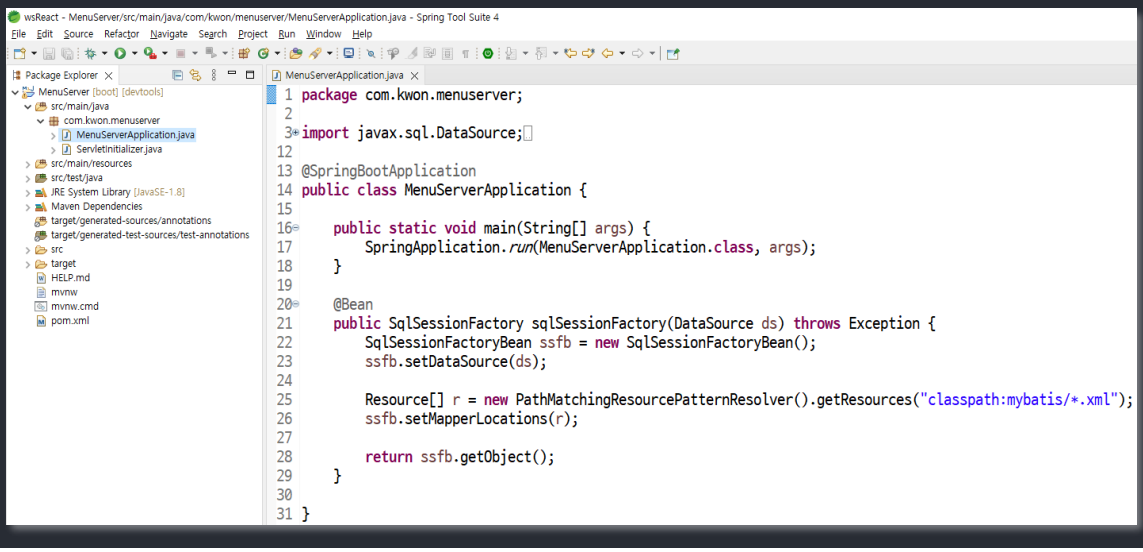
```sql
create table menu_react_kwon(
    m_name varchar2(10 char) primary key,
    m_price number(5) not null
);

select * from menu_react_kwon;

insert into menu_react_kwon values('야채김밥', 4000);
```

1.2.5 mapper.xml 들 위치 등록

MenuServerApplication.java 에



1.3 Spring Boot + MyBatis

1.3.1 JavaBean

Menu.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/Menu.java - Spring Tool Suite 4

```java
package com.kwon.menuserver.menu;

import java.math.BigDecimal;

public class Menu {
    private String m_name;
    private BigDecimal m_price;

    public Menu() {
        // TODO Auto-generated constructor stub
    }

    public Menu(String m_name, BigDecimal m_price) {
        super();
        this.m_name = m_name;
        this.m_price = m_price;
    }

    public String getM_name() {
        return m_name;
    }

    public void setM_name(String m_name) {
        this.m_name = m_name;
```

1.3.2 mapper.xml

src/main/resources/mybatis 폴더 만들고

menuMapper.xml

wsReact - MenuServer/src/main/resources/mybatis/menuMapper.xml - Spring Tool Suite 4

```xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE mapper PUBLIC "-//mybatis.org//DTD Mapper 3.0//EN" "http://
<mapper namespace="com.kwon.menuserver.menu.MenuMapper">
    <select id="getMenu" resultType="com.kwon.menuserver.menu.Menu">
        select * from menu_react_kwon
    </select>
</mapper>
```

1.3.3 Mapper

MenuMapper.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/MenuMapper.java - Spring Tool Suite 4

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- MenuServer [boot] [devtools]
  - src/main/java
    - com.kwon.menuserver
    - com.kwon.menuserver.menu
      - Menu.java
      - MenuMapper.java
  - src/main/resources
    - mybatis
      - menuMapper.xml
    - static
    - templates
    - application.properties
  - src/test/java
  - JRE System Library [JavaSE-1.8]

Menu.java | menuMapper.xml | MenuMapper.java

```java
package com.kwon.menuserver.menu;

import java.util.List;

import org.apache.ibatis.annotations.Mapper;

@Mapper
public interface MenuMapper {
    public List<Menu> getMenu();
}
```

1.3.4 DAO

MenuDAO.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/MenuDAO.java - Spring Tool Suite 4

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- MenuServer [boot] [devtools]
  - src/main/java
    - com.kwon.menuserver
    - com.kwon.menuserver.menu
      - Menu.java
      - MenuDAO.java
      - MenuMapper.java
  - src/main/resources
  - src/test/java
  - JRE System Library [JavaSE-1.8]
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations
  - src
  - target
  - HELP.md
  - mvnw
  - mvnw.cmd
  - pom.xml

MenuDAO.java

```java
package com.kwon.menuserver.menu;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

@Service
public class MenuDAO {

    @Autowired
    private MenuMapper mm;

    public void getMenu() {
        List<Menu> menus = mm.getMenu();
        for (Menu m : menus) {
            System.out.println(m.getM_name());
            System.out.println(m.getM_price());
        }
    }
}
```

1.3.5 Controller

MenuController.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/MenuController.java - Spring Tool Suite 4

```java
package com.kwon.menuserver.menu;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
import org.springframework.web.bind.annotation.ResponseBody;

@Controller
public class MenuController {

    @Autowired
    private MenuDAO mDAO;

    @RequestMapping(value = "/menu.get", method = RequestMethod.GET)
    public @ResponseBody String menuGet() {
        mDAO.getMenu();
        return "우리가 90 나왔다는데 뭐";
    }
}
```

1.4 기능 구현

1.4.1 라이브러리 추가

pom.xml 에

wsReact - MenuServer/pom.xml - Spring Tool Suite 4

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="ht
    <modelVersion>4.0.0</modelVersion>
    <parent>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-parent</artifactId>
        <version>2.7.5</version>
        <relativePath /> <!-- lookup parent from repository -->
    </parent>
    <groupId>com.kwon</groupId>
    <artifactId>MenuServer</artifactId>
    <version>0.0.1-SNAPSHOT</version>
    <packaging>war</packaging>
    <name>MenuServer</name>
    <description>Demo project for Spring Boot</description>
    <properties>
        <java.version>1.8</java.version>
    </properties>
    <dependencies>

        <dependency>
            <groupId>org.codehaus.jackson</groupId>
            <artifactId>jackson-mapper-asl</artifactId>
            <version>1.9.13</version>
        </dependency>

        <dependency>
```

1.4.2 JavaBean

Menus.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/Menus.java - Spring Tool Suite 4

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- MenuServer [boot] [devtools]
  - src/main/java
    - com.kwon.menuserver
    - com.kwon.menuserver.menu
      - Menu.java
      - MenuController.java
      - MenuDAO.java
      - MenuMapper.java
      - Menus.java
  - src/main/resources
  - src/test/java
  - JRE System Library [JavaSE-1.8]
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations
  - src
  - target
  - HELP.md
  - mvnw
  - mvnw.cmd
  - pom.xml

Menus.java

```java
package com.kwon.menuserver.menu;

import java.util.List;

public class Menus {

    private List<Menu> menu;

    public Menus() {
        // TODO Auto-generated constructor stub
    }

    public Menus(List<Menu> menu) {
        super();
        this.menu = menu;
    }

    public List<Menu> getMenu() {
        return menu;
    }

    public void setMenu(List<Menu> menu) {
        this.menu = menu;
    }

}
```

1.4.3 mapper.xml

menuMapper.xml

wsReact - MenuServer/src/main/resources/mybatis/menuMapper.xml - Spring Tool Suite 4

File Edit Source Refactor Source Navigate Search Project Run Window Help

Package Explorer

- MenuServer [boot] [devtools]
  - src/main/java
  - src/main/resources
    - mybatis
      - menuMapper.xml
    - static
    - templates
    - application.properties
  - src/test/java
  - JRE System Library [JavaSE-1.8]
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations

menuMapper.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE mapper PUBLIC "-//mybatis.org//DTD Mapper 3.0//EN" "http://my
<mapper namespace="com.kwon.menuserver.menu.MenuMapper">
    <select id="getMenu" resultType="com.kwon.menuserver.menu.Menu">
        select * from menu_react_kwon order by m_name
    </select>
    <insert id="regMenu" parameterType="com.kwon.menuserver.menu.Menu">
        insert into menu_react_kwon values(#{m_name}, #{m_price})
    </insert>
</mapper>
```

1.4.4 Mapper

MenuMapper.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/MenuMapper.java - Spring Tool Suite 4

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer

- MenuServer [boot] [devtools]
  - src/main/java
    - com.kwon.menuserver
    - com.kwon.menuserver.menu
      - Menu.java
      - MenuController.java
      - MenuDAO.java
      - MenuMapper.java
      - Menus.java
  - src/main/resources
  - src/test/java
  - JRE System Library [JavaSE-1.8]

MenuMapper.java

```java
package com.kwon.menuserver.menu;

import java.util.List;

@Mapper
public interface MenuMapper {
    public List<Menu> getMenu();
    public int regMenu(Menu m);
}
```

1.4.5 DAO

MenuDAO.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/MenuDAO.java - Spring Tool Suite 4

```java
package com.kwon.menuserver.menu;

import java.util.ArrayList;

@Service
public class MenuDAO {

    @Autowired
    private MenuMapper mm;

    public Menus getMenu() {
        try {
            return new Menus(mm.getMenu());
        } catch (Exception e) {
            e.printStackTrace();
            return new Menus(new ArrayList<>());
        }
    }

    public Menu regMenu(Menu m) {
        try {
            if (mm.regMenu(m) == 1) {
                return m;
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return new Menu(null, null);
    }
}
```

1.4.6 Controller

MenuController.java

wsReact - MenuServer/src/main/java/com/kwon/menuserver/menu/MenuController.java - Spring Tool Suite 4

```java
package com.kwon.menuserver.menu;

import javax.servlet.http.HttpServletResponse;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
import org.springframework.web.bind.annotation.ResponseBody;

@Controller
public class MenuController {

    @Autowired
    private MenuDAO mDAO;

    @RequestMapping(value = "/menu.get", method = RequestMethod.GET, produces = "application/json; charset=utf-8")
    public @ResponseBody Menus menuGet(HttpServletResponse res) {
        res.setHeader("Access-Control-Allow-Origin", "*");
        return mDAO.getMenu();
    }

    @RequestMapping(value = "/menu.reg", method = RequestMethod.GET, produces = "application/json; charset=utf-8")
    public @ResponseBody Menu menuReg(Menu m, HttpServletResponse res) {
        res.setHeader("Access-Control-Allow-Origin", "*");
        return mDAO.regMenu(m);
    }
}
```

2. React Front-end

2.1 일단

프로젝트 만들고 axios 넣고

2.2 조회

menu.js

```
import axios from "axios";
import React, { useEffect, useState } from "react";

const Menu = () => {
  const [menus, setMenus] = useState([]);

  const getMenu = () => {
    let ms = [];
    axios.get("http://192.168.0.125/MS/menu.get").then((res) => {
      res.data.menu.forEach((m) => {
        ms = ms.concat(
          <tr>
            <td>{m.m_name}</td>
            <td>{m.m_price}</td>
          </tr>
        );
      });
      setMenus(ms);
    });
  };

  useEffect(() => {
    getMenu();
  }, []);

  return (
    <>
      <table border={1}>
        <tr>
          <th>메뉴명</th>
          <th>가격</th>
        </tr>
        {menus}
      </table>
    </>
  );
};

export default Menu;
```

App.js

```
import "./App.css";
import Menu from "./menu";

function App() {
  return (
    <>
      <Menu />
    </>
  );
}

export default App;
```

2.3 추가

menu.js

```
import axios from "axios";
import React, { useEffect, useState } from "react";

const Menu = () => {
  const [menus, setMenus] = useState([]);
  const [m, setM] = useState({ m_name: "", m_price: "" });

  const getMenu = () => {…
  };
  useEffect(() => {…
  }, []);

  const updateM = (e) => {
    setM({ ...m, [e.target.name]: e.target.value });
  };

  const reg = () => {
    axios.get("http://192.168.0.125/MS/menu.reg", { params: m }).then((res) => {
      alert(res.data.m_name);
      alert(res.data.m_price);
      getMenu();
      setM({ m_name: "", m_price: "" })
    });
  };
  return (
    <>
      메뉴명 : <input name="m_name" value={m.m_name} onChange={updateM} /><br />
      가격 : <input name="m_price" value={m.m_price} onChange={updateM} /><br />
      <button onClick={reg}>등록</button>
      <hr />
      <table border={1}>
        <tr><th>메뉴명</th><th>가격</th></tr>
        {menus}
      </table>
    </>
  );
};

export default Menu;
```